Zhenghao He

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Research Interests

- **Reasoning in foundation models:** understanding and improving multi-step reasoning, test-time computation, and inference-time scaling of large language models.
- **Efficient inference:** faster and more reliable decoding for large language models, including diffusion LMs.
- **Interpretability and controllability:** mechanistic analysis of internal representations and representation-level steering to make model behavior transparent and controllable.

Education

University of Virginia — Charlottesville, VA, USA
Ph.D. in Computer Science (GPA: 3.9/4.0); Advisor: Prof. Aidong Zhang — *08/2024 – 05/2029 (expected)*

- Available for a 12-week internship: 05/2027 – 08/2027

Tongji University — Shanghai, China
B.E. in Computer Science and Technology (GPA: 88.5/100) — *09/2020 – 07/2024*

- Relevant coursework: Machine Learning, Deep Learning, Data Mining, Computer Vision, Algorithms

Publications

LLM Reasoning, Post-Training, and Efficient Inference

[arXiv 2026] Zhenghao He, Bohan Liu, Guangzhi Xiong, Aidong Zhang. *Reliable Parallel Decoding in Masked Diffusion Language Models.* [arXiv] [Code] [Website]
▷ Training-free parallel decoding, 2.4–6.1× faster than default decoding on LLaDA and Dream at comparable or better accuracy.

[arXiv 2026] Zhenghao He, Guangzhi Xiong, Sanchit Sinha, Bohan Liu, Wenqian Ye, Aidong Zhang. *Rethinking Reasoning Paths as Phase-Structured Trajectories.* [arXiv]
▷ Phase-aligned probing of reasoning trajectories that improves Best-of-N selection and enables phase-wise steering.

[EMNLP 2026 Main] Zhenghao He, Guangzhi Xiong, Bohan Liu, Sanchit Sinha, Aidong Zhang. *Triggering Chain-of-Thought via Latent Feature Interventions in Large Language Models.* [arXiv] [Code]
▷ Steering a few SAE features induces reasoning without CoT prompting, matching CoT accuracy across model families.

[NeurIPS 2026] Guangzhi Xiong, **Zhenghao He**, Bohan Liu, Sanchit Sinha, Wenqian Ye, Aidong Zhang. *Capturing In-Context Learning Dynamics with Task Operators.* [arXiv] [Code]
▷ Models attention heads as affine maps of context, enabling ICL compression and many-shot scaling without longer contexts.

[Under Review] Wenqian Ye, Bo Yuan, Bohan Liu, Guangzhi Xiong, **Zhenghao He**, Yongxin Chen, Henry Kautz, Aidong Zhang. *Demystifying and Accelerating Emergence to New Tasks in Reinforcement Learning.*
▷ Circuit-level analysis of delayed generalization in RLVR; a spectral intervention speeds emergence under PPO/GRPO on Qwen3.

Interpretability and Trustworthy AI

[NeurIPS 2026] Zhenghao He, Guangzhi Xiong, Boyang Wang, Sanchit Sinha, Aidong Zhang. *CASL: Concept-Aligned Sparse Latents for Interpreting Diffusion Models.* [arXiv]
▷ First supervised alignment between SAE latents and semantic concepts in diffusion models, enabling precise concept-level control.

[ICCV 2025] Zhenghao He, Sanchit Sinha, Guangzhi Xiong, Aidong Zhang. *GCAV: A Global Concept Activation Vector Framework for Cross-Layer Consistency in Interpretability.* [arXiv] [Code]
▷ Unifies concept activation vectors across layers via contrastive alignment for stable concept attributions.

[ICLR 2026] Guangzhi Xiong, **Zhenghao He**, Bohan Liu, Sanchit Sinha, Aidong Zhang. *Toward Faithful Retrieval-Augmented Generation with Sparse Autoencoders.* [arXiv] [Code]
▷ Lightweight SAE-based detector that flags unfaithful RAG outputs from LLM internals with interpretable rationales.

[ECCV 2026] Bohan Liu, Wenqian Ye, Guangzhi Xiong, **Zhenghao He**, Sanchit Sinha, Aidong Zhang. *Towards Robustness against Typographic Attack with Training-free Concept Localization.* [arXiv] [Code]
▷ Training-free intervention on lexically biased attention heads in CLIP, improving LVLM robustness to typographic attacks.

[Under Review] Bohan Liu, **Zhenghao He**, Wenqian Ye, Guangzhi Xiong, Sanchit Sinha, Aidong Zhang. *FlowScope: Interpreting Generation Dynamics in Continuous Flow Language Models.*

[Under Review] Bohan Liu, Wenqian Ye, Guangzhi Xiong, **Zhenghao He**, Sanchit Sinha, Aidong Zhang. *Label-Free Localization and Test-Time Suppression of CLIP's Typographic Circuits.*

Multimodal Reasoning

[CVPR 2026] Guangzhi Xiong, Sanchit Sinha, **Zhenghao He**, Aidong Zhang. *Retrieving Counterfactuals Improves Visual In-Context Learning.* [arXiv] [Code]

▷ Retrieves counterfactual-style demonstrations via composed image retrieval, improving VLM in-context reasoning and robustness.

[AAAI 2026, Oral] Sanchit Sinha, Guangzhi Xiong, **Zhenghao He**, Aidong Zhang. *Concept-RuleNet: Grounded Multi-Agent Neurosymbolic Reasoning in Vision Language Models.* [arXiv]

▷ Grounded neurosymbolic VLM reasoning that improves accuracy by 5% on average and cuts hallucinated symbols by up to 50%.

[Under Review] Sanchit Sinha, Guangzhi Xiong, Bohan Liu, **Zhenghao He**, Aidong Zhang. *Medic-MAS: Mitigating Visual Forgetting in Latent Multi-Agent Reasoning for Medical Visual Question Answering.*

Earlier Undergraduate Work

[ICSCA 2023] **Zhenghao He**, Ruifan Chen, Yayue Hou, Fei Xie, Xiaoliang Gong, Anthony G. Cohn. *A Music Labeling Model Based on Traditional Chinese Music Characteristics for Emotional Regulation.*

[ICSCA 2023] Fei Xie, Xiaoliang Gong, **Zhenghao He**, Tongqi Wu, Yan Lu, Mohan Zhao. *Alzheimer's Disease Early Diagnosis Based on Resting-State Dynamic Functional Connectivity.*

[MICCAI Workshop 2022] Mohan Zhao, Lu Pang, Yan Lu, Fei Xie, **Zhenghao He**, Xiaoliang Gong, Anthony G. Cohn. *Conditional Domain Adaptation Based on Initial Distribution Discrepancy for EEG Emotion Recognition.*

[MLISE 2022] **Zhenghao He**. *Comparison of Different Machine Learning Methods Applied to Obesity Classification.*

Work Experience

Alibaba Cloud — Shanghai, China
AI Model Optimization Intern — *03/2024 – 08/2024*

- Surveyed GPU failure prediction and anomaly detection methods for large-scale distributed training clusters.
- Analyzed cluster telemetry (utilization, temperature, memory) to identify potential indicators of GPU failures.

HaaenClean Group, Digital Center — Shanghai, China
Digital Engineer Intern — *09/2023 – 12/2023*

- Built and deployed an anomaly detection pipeline for multi-sensor industrial data.
- Developed an on-premise LLM-based system for secure document retrieval and automated report generation.

Shanghai Research Institute for Intelligent Autonomous Systems — Shanghai, China
Research Assistant to Assoc. Prof. Zhongpan Zhu, Tongji University — *06/2022 – 09/2022*

- Proposed a patent, "Algorithm of Generating Heat Map for Driver's Attention," for vision-based driver monitoring.

Teaching & Service

- **Teaching Assistant**, CS 6316 Machine Learning, University of Virginia — Fall 2026
- **Conference Reviewer**: NeurIPS, ICLR, ACL Rolling Review (ARR)

Honors & Awards

- UVA Computer Science Scholarship — 08/2024
- Excellent Student Scholarship, Tongji University — 12/2023
- 2nd Prize (National Level), Intel Cup Embedded System Design Invitational Contest — 08/2022

Skills

- **Programming:** Python, C/C++, CUDA, MATLAB
- **Frameworks & Tools:** PyTorch, TensorFlow, Git, Linux, LaTeX
- **Languages:** Chinese (native), English (advanced)